

CrewAI Multi-Agent Collaboration

Week 2 Day 4 - Internship Assignment

1. Overview

This report documents the implementation of a multi-agent system using CrewAI to solve a business task: researching a competitor (Canva), analyzing findings, and drafting a marketing angle.

2. Business Task

Objective

Research Canva (the design tool company), summarize findings into a SWOT analysis, and draft 2-3 marketing angles for competing against them.

Why Multi-Agent?

The task has three distinct phases requiring different expertise:

- Research - factual data gathering (needs temp=0, no creativity)
- Analysis - synthesizing insights (needs temp=0.3, structured)
- Strategy - creative marketing angles (needs temp=0.7, creative)

Each phase benefits from a specialized agent with a focused persona and appropriate temperature setting.

3. Implementation

Agents Defined

Agent	Role	Goal	Temp
Researcher	Competitor Research Specialist	Gather factual data	0.0
Analyst	Competitive Insights Analyst	Synthesize insights	0.3
Strategist	Marketing Angle Strategist	Create marketing angles	0.7

Tools Assigned

Agent	Tools	Justification
Researcher	web_search	Needs to find data
Analyst	write_file	Needs to save analysis
Strategist	write_file	Needs to save angles

Process Used

Sequential: Agents run in order, output flows from one to the next.

Researcher -> Analyst -> Strategist -> Final Result

Hierarchical: Manager agent oversees work, reviews output, can request revisions.

Manager -> Researcher -> Review -> Analyst -> Review -> Strategist -> Review -> Final Result

4. Results

Sequential Run

The researcher produced a structured Canva report with 4 sections (Products, Pricing, Market Position, Recent News). The analyst created a SWOT analysis with 3 key insights. The strategist drafted 3 marketing angles and recommended 'Design for the Future' as the strongest angle.

Hierarchical Run

The manager oversaw the same workflow but reviewed each agent's output before proceeding. The final output was slightly more consistent in formatting.

Token Usage and Cost

Approach	Tokens	Cost	Quality
Sequential	~5,000	~\$0.0014	4.67/5
Hierarchical	~7,000	~\$0.0019	5.00/5

Cost calculation: OpenRouter Llama 3.3 70B at ~\$0.27 per million tokens.

Success Criteria

Criterion	Sequential	Hierarchical
Factual grounding	4/5	5/5
Completeness	5/5	5/5
Tone	5/5	5/5
Average	4.67/5	5.00/5

5. Sequential vs Hierarchical

Feature	Sequential	Hierarchical
Speed	Faster	Slower (reviews)
Quality	Good	Better
Cost	Lower	Higher (+40%)
Reliability	Predictable	Depends on manager
Error handling	No correction	Manager can redo

6. Verdict

The multi-agent crew was worth the added complexity for this task. The hierarchical crew produced slightly better output because the manager caught formatting inconsistencies and ensured each agent followed the expected output format. However, it cost ~40% more tokens.

Recommendation: Start with sequential for simple tasks. Switch to hierarchical only when you need quality assurance or when the task is too complex for agents to handle independently.

7. Technical Challenges Resolved

- cache_breakpoint error - CrewAI sends cache_breakpoint, Groq/OpenRouter don't support it. Fixed with monkey-patch.
- Experimental executor - CrewAI 1.15.x adds built-in tools causing errors. Fixed with executor_type='crew'.
- Jupyter async conflict - crew.kickoff() fails in Jupyter. Fixed by running as .py script.

- Groq rate limits - Free tier (8000 TPM) too low. Fixed by using OpenRouter.
- Windows encoding - CrewAI outputs contain unicode. Fixed with UTF-8 encoding.

8. Files

File	Purpose
task3_runner.py	Sequential crew execution
task4_runner.py	Hierarchical crew execution
task5_runner.py	Evaluation and cost comparison
notes.md	Detailed notes for all 5 tasks
writeup.md	Write-up (markdown)
.env	API keys (Groq, OpenRouter)